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A comparative study of dimensionality reduction methods for large-scale image retrieval

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ABSTRACT

“Curse of Dimensionality” is one of the important problems that Content-Based Image Retrieval (CBIR) confronts. Dimensionality reduction is an effective method to overcome it. In this paper, six commonly-used dimensionality reduction methods are compared and analyzed to examine their respective performance in image retrieval. The six methods include Principal Component Analysis (PCA), Fisher Linear Discriminant Analysis (FLDA), Local Fisher Discriminant Analysis (LFDA), Isometric Mapping (ISOMAP), Locally Linear Embedding (LLE), and Locality Preserving Projections (LPP). For comparison, Scale Invariant Feature Transform (SIFT) and color histogram in Hue, Saturation, Value (HSV) color space are firstly extracted as image features, meanwhile SIFT feature extraction procedure is optimized to reduce the number of SIFT features. Then, PCA, FLDA, LFDA, ISOMAP, LLE, and LPP are respectively applied to reduce the dimensions of feature vectors, which can be used to generate vocabulary trees. Finally, we can process large-scale image retrieval based on the inverted index built by vocabulary trees. In the experiments, the performance of various dimensionality reduction methods are analyzed comprehensively by comparing the retrieval performance, advantages and disadvantages, computational complexity and time-consuming of image retrieval. Through a series of experiments, we can conclude that dimensionality reduction method of LLE and LPP can effectively reduce computational complexity of image retrieval, while maintaining high retrieval performance.

1. Introduction

With the rapid development of the Internet, the popularity of various digital equipment, increasing mass storage devices, and the number of digital images have growth exponentially. It is important on how to effectively manage and rationally utilize such large-scale image information, which has become a challenging and urgent problem in the field of multimedia information retrieval [1–3].

Since the 1990 s, Content-Based Image Retrieval (CBIR) has appeared and flourished gradually, and become a mainstream image retrieval method nowadays [4–6]. The image features are used to represent the content of images. The similarity between images is measured and the images that have similar features with the query image will be regarded as the search results. With specific image feature extraction algorithms, extracted image features are invariant and objective, and require no manual operation, which enable CBIR as a retrieval method to reflect image content automatically and objectively.

The key technologies of CBIR include feature extraction, similarity measurement, and relevance feedback, etc. The most critical parts are feature extraction and similarity measurement. For CBIR, image features are used to represent the image content, therefore the extracted image features will directly have effects on performance of image retrieval. The features extracted from images can be defined into two levels: low-level features which refer to one aspect of physical characteristics; while high-level features are semantic characteristics which can reflect semantic level and understanding of image content. Therefore, how to extract image features rapidly and efficiently is a key problem of CBIR.

The existing CBIR system usually adopts visual features to represent image content, such as color, texture, shape, contour and spatial relationships, etc. High dimension features are usually extracted to describe image content accurately. Especially for large-scale image retrieval system, the number of features is large-scale as well. These high-dimensional image features may lead to “Curse of Dimensionality” which will cause a degraded retrieval performance. The primary reason of “Curse of Dimensionality” is that the distribution of original data is extremely sparse. Large deviation will appear in similarity measure of image features, which will result in inefficient performance of processing algorithm of features. Correspondingly, dimensionality reduction is an effective method to overcome this problem. Through this

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method, image features are preprocessed by projecting original data from a high-dimensional space to a lower dimensional space. In brief, dimensionality reduction method is an important step for CBIR.

The existing dimensionality reduction methods can be roughly divided into two categories: the unsupervised and the supervised. Unsupervised dimensionality reduction can minimize the loss of data information. The commonly used unsupervised dimensionality reduction methods include Principal Component Analysis (PCA) [7], Multidimensional Scaling (MDS) [8] and kernel PCA (KPCA) [9]. PCA aims to find the optimal projection matrix in terms of least mean square. MDS makes the Euclidean distance between data consistent with the distance of original data after dimension reduction. KPCA is an improved variety of PCA. By giving appropriate kernel function, PCA reconstruction in kernel space becomes relatively simple to handle nonlinear data. On the other hand, supervised methods can maximize inter-class discriminant information. The two commonly used supervised methods include Fisher Linear Discriminant Analysis (FLDA) [10], and Local Fisher Discriminant Analysis (LFDA) [11].

In addition to the dimensionality reduction methods mentioned above, manifold learning [12,13] is a novel method occurred in recent years, which learns and discovers smooth low-dimensional manifold embedded in high-dimensional space of finite discrete samples. It reveals low-dimensional structure in high-dimensional data, and then reconstructs and performs non-linear dimensionality reduction. The essence of manifold learning is a kind of nonlinear dimensionality reduction that can find intrinsic geometric constructions or rules while sample space is smooth high-dimensional, and obtain corresponding low-dimensional embedding. This method is better than traditional methods of dimensionality reduction in terms of reflecting the essence of data, and is more conducive in the understanding of data and further processing. The representative methods of the manifold learning are Isometric Mapping (ISOMAP) [14,15], Locally Linear Embedding (LLE) [16], and Locality Preserving Projections (LPP) [17] as well.

In this paper, six dimensionality reduction methods are compared and analyzed comprehensively to demonstrate their own performances in image retrieval. For effective comparison, Scale Invariant Feature Transform (SIFT) features [18] and color histogram in Hue, Saturation, Value (HSV) [19] color space are firstly extracted as image features, and SIFT feature extraction procedure is optimized to reduce the number of SIFT features. Then, PCA, FLDA, LFDA, ISOMAP, LLE, and LPP are respectively applied to reduce the dimensions of the image features, which can be used to generate vocabulary trees. Finally, through building an inverted index of vocabulary trees, large-scale image retrieval is implemented. In the experiments, the performance of various dimensionality reduction methods are analyzed by comparing the retrieval performance, advantages and disadvantages [20], computational complexity and time-consuming of image retrieval. It can be concluded that LLE and LPP methods can effectively reduce the computational complexity of image retrieval, while maintaining the high retrieval performance.

The remainder of this paper is organized as follows: Section 2 introduces the large-scale image retrieval framework based on vocabulary tree adopted in this paper, and a series of comparisons between the six dimensionality reduction methods. Based on this framework, the performance of various dimensionality reduction methods on image retrieval is compared. Section 3 presents the experimental results. Conclusions are drawn in Section 4.

2. Large-scale image retrieval framework based on vocabulary tree

In this paper, a large-scale image retrieval framework based on vocabulary tree [21] is firstly set up for a performance comparison of the six dimensionality reduction methods. The retrieval process is shown in (Fig. 1. First, image features are extracted from image database. Then, the dimension of these features is reduced and then used to construct two vocabulary trees by means of hierarchical K-means clustering scheme. By counting inverted index [22] based on vocabulary tree, the features and their indexes are stored in the image database.

When operating a query, the features of query image are extracted and indexed using the vocabulary tree. The similarity of images is matched and ranked by comparing the index of the query image with the indexes that are stored in the database. Finally, the top-k images are returned to the users.

As Fig. 1 shows, the image retrieval framework includes four key segments: feature extraction, dimensionality reduction, the construction of image index based on visual vocabulary tree and similarity matching, which will be further introduced below.

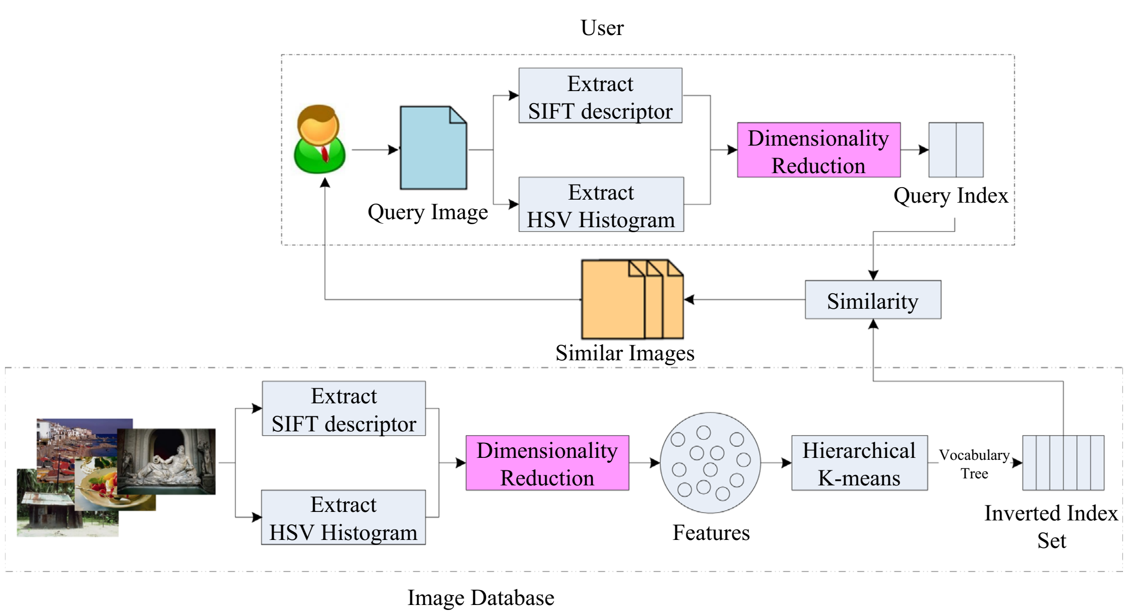

Fig. 1. The framework of large-scale image retrieval based on vocabulary tree.

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2.1. Feature extraction

It is clear that feature extraction is one of the most critical parts of the image retrieval framework. In this paper, color and SIFT features are extracted as the image features. Plus, SIFT extraction procedure is optimized to reduce the number of SIFT features.

2.1.1. SIFT feature

The traditional SIFT feature extraction procedure is shown in Fig. 2(a), where the crossings present the coordinates of SIFT features. It can be found that SIFT descriptors are over-concentrated in the areas with similar characteristics. Therefore, we propose a method to optimize SIFT feature extraction procedure so as to reduce the number of SIFT features. After optimization, the image content can be still characterized accurately, but with fewer SIFT descriptors.

Suppose that $Sift_{des}[i].x$, $Sift_{des}[i].y$ respectively represents the horizontal and vertical coordinates of i-th SIFT; T_{opt} is set as an optimization threshold and R_{opt} as optimization range. For any two different SIFT descriptors $Sift_{des}[i]$ and $Sift_{des}[j]$, when the distance of horizontal and vertical coordinates of two points are less than optimal threshold T_{opt}, these points can be merged into one because they exist in the optimization range R_{opt}. Otherwise, it is needless to optimize. The optimization process can be represented in Eq. (1):

$$\underset{i \neq j}{Sift_{des}[i], Sift_{des}[j]} \begin{cases} \in R_{opt}, & if\ ||Sift_{des}[i]x - Sift_{des}[j]x|| \leq T_{opt} \\ \&\&||Sift_{des}[i]y - Sift_{des}[j]y|| \leq T_{opt} & \\ \notin R_{opt}, & \text{otherwise} \end{cases} \tag{1}$$

The effect of optimized SIFT is shown in Fig. 2(b). Compared with the Fig. 2(a), the number of the optimized SIFT feature points have been reduced remarkably. And the experimental results show that the optimization has little influence on the retrieval performance [23].

2.1.2. Color feature

Since SIFT features merely exploit gray information, fault detection may happen when images share similar outline while colors are different. To overcome this shortage, we combine the color features with SIFT to further improve the retrieval performance.

Firstly, the color features are extracted in HSV color space. H, S, V are divided into 8, 3, 3 bins respectively through the non-interval quantization, which are represented as H', S', V'. According to Eq. (2), each one-dimensional vector consists of 72 bins. Color histogram formed by the vectors is used as the color features.

$$I = H'Q_S Q_V + S'Q_V + V' \tag{2}$$

where Q_S, Q_V are the quantization levels of S', V', Q_S=3, Q_V=3.

2.2. Dimensionality reduction

As mentioned above, SIFT features and color histogram are extracted as image features to represent the image content. For a large-scale image retrieval system, the number of images and features are vast as well. Because that the high-dimensional features contain a large number of redundant information and there are certain correlations hiding among feature data, it may lead to “Curse of Dimensionality”. Thus, the original features will be preprocessed with dimensionality reduction methods before the similarity matching, and the high-dimensional features are simplified by finding low-dimensional structure within them. Although the performance of image retrieval will be decreased slightly while reducing the amount of features to be processed, it can eliminate some redundant information to achieve the reduction of computational complexity and reach the purpose of improving the speed of image retrieval significantly.

Assume that eigenvectors is $X \in R^{D \times N}$, N is the total number of features, and D is the dimension of each eigenvector. Dimensionality Reduction methods utilize the projection matrix $W = \{w_i\} \in R^{D \times d}$ ($d < D$) to reduce the dimension of features, and the low-dimensional features embedded in the feature space can be found:

$$Y = W^T X \tag{3}$$

As Fig. 3 shown, dimensionality reduction methods can be classified into the unsupervised and the supervised ones. The unsupervised and supervised dimensionality reduction methods to be compared in this paper include PCA, ISOMAP, LLE, LPP and FLDA, LFDA as well. They will be further introduced below.

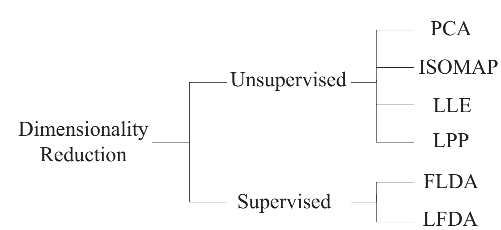

Fig. 3. The classification of dimensionality reduction methods in this paper.

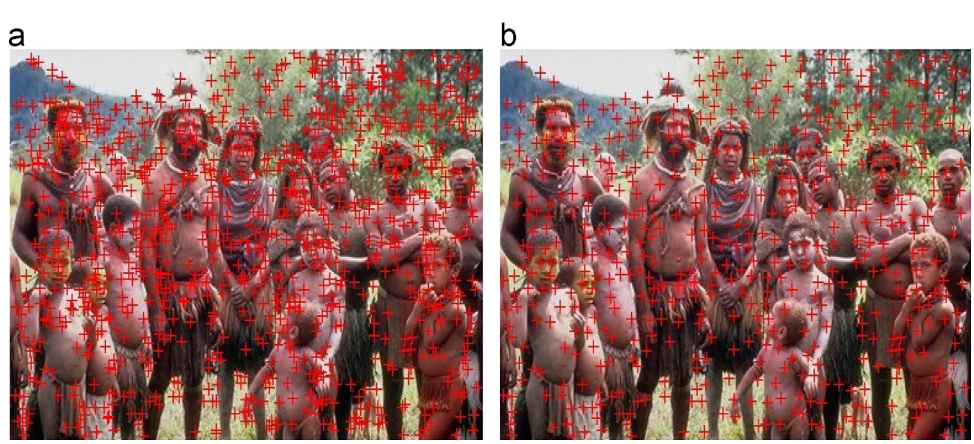

Fig. 2. SIFT descriptors. (a) Traditional SIFT. (b) Optimized SIFT.

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2.2.1. PCA

PCA is a kind of unsupervised linear dimensionality reduction method in which the data space is projected to a low-dimensional space by orthogonal transformation.

First, the covariance matrix is computed as in Eqs. (4) and (5):

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{4}$$

$$Cov = \sum_{i=1}^{N} (x_i-\mu)(x_i-\mu)^T \tag{5}$$

where N features are denoted as $\{x_1, x_2,\ldots, x_N\}$, μ is mean of all image features, and Cov is covariance matrix.

Second, after the liner transformation V^T, the eigenvector of W_{PCA} is the optimal projection matrix in the sense of MMSE and it is computed as in Eq. (6):

$$W_{PCA} = \arg\max_V |V^T Cov V| \tag{6}$$

Finally, Eq. (3) is used for dimensionality reduction.

2.2.2. FLDA

FLDA is a supervised linear dimensionality reduction method. The projection matrix W_{flda} is obtained by maximizing the inter-class scatter matrix S_{inter} while minimizing the intra-class scatter matrix S_{intra} as in Eq. (7):

$$W_{flda} = \arg\max_V \left|\frac{V^T S_{int\,er} V}{V^T S_{int\,ra} V}\right| \tag{7}$$

where V^T is the liner transformation matrix,

$$S_{int\,er} = \frac{1}{2}\sum_{i,j=1}^{N} Inter_{i,j}(x_i-x_j)(x_i-x_j)^T \tag{8}$$

$$S_{int\,ra} = \frac{1}{2}\sum_{i,j=1}^{N} Intra_{i,j}(x_i-x_j)(x_i-x_j)^T \tag{9}$$

$$Inter_{i,j} = \begin{cases} \frac{1}{n_c} i,j \in C \\ 0 \ when \ i \in C, j \notin C \end{cases} \tag{10}$$

$$Intra_{i,j} = \begin{cases} (\frac{1}{N}-\frac{1}{n_c}) \ \ i,j \in C \\ \frac{1}{N} \ \ when \ \ i \in C, j \notin C \end{cases} \tag{11}$$

In general, the number of extracted features is much more than the number of images so that the resulting intra-class scatter matrix S_{intra} is singular which makes the eigenvector matrix unsolvable. In order to overcome this problem, PCA is usually used to reduce the dimension of the input features so that S_{intra} becomes non-singular. W_{FLDA} is calculated as in Eqs. (12) and (13):

$$W_{FLDA} = W_{PCA} W_{flda} \tag{12}$$

$$W_{flda} = \arg\max_V \left|\frac{V^T W_{PCA}^T S_{int\,er} W_{PCA} V}{V^T W_{PCA}^T S_{int\,ra} W_{PCA} V}\right| \tag{13}$$

FLDA preserves the discriminative of image features while reducing dimension in the high-dimensional space. Nevertheless, the intra-class scatter S_{intra} is always singular when image features in a particular class have many local means, which prevent it from being correct.

2.2.3. LFDA

LFDA is the extension of FLDA. The projection matrix W_{lfda} consists of maximizing the local inter-class scatter matrix V_{inter} and minimizing the local intra-class scatter matrix V_{intra}. Unlike FLDA, the parameters of $inter_{i,j}$ and $intra_{i,j}$ can be calculated as in Eqs. (14–16):

$$Inter_{i,j} = \begin{cases} \frac{A_{i,j}}{n_c} \ \ i,j \in C \\ 0 \ \ when \ \ i \in C, j \notin C \end{cases} \tag{14}$$

$$Intra_{i,j} = \begin{cases} A_{i,j}\left(\frac{1}{N}-\frac{1}{n_c}\right) \ \ i,j \in C \\ \frac{1}{N} \ \ when \ \ i \in C, j \notin C \end{cases} \tag{15}$$

$$A_{i,j} = \frac{e^{(-\|x_i-x_j\|_2^2)}}{\rho_i \rho_j} \tag{16}$$

where N is the total number of images, C denotes a certain kind of classes, and n_c is the number of the Cth class. $\rho_i = \|x_i - x_i^k\|_2$, x_i^k is the k-nearest-neighbors of x_i [24], and k is the tuning factor. Similarly, PCA is used to reduce the dimension of the input features so that the V_{intra} becomes non-singular:

$$W_{LFDA} = W_{PCA} W_{lfda} \tag{17}$$

It makes the optimal effect working for both inter-class classification and intra-class.

2.2.4. ISOMAP

Based on MDS, ISOMAP uses geodesic distance to replace Euclidean distance. The ISOMAP algorithm includes three steps:

(1) Construct the adjacency graph, and compute the k-nearest neighbors of each point in the graph;
(2) Compute the shortest distances to approximate the geodesic distances. The algorithm of *Dijkstra* [25] can be used to compute the shortest distances corresponding the geodesic distance as approximation of the manifold;
(3) Let the distances as input data and construct an embedding low-dimension space to achieve the dimensionality reduction.

There are two parameters in ISOMAP: the size of k-nearest neighbors and the dimension of reduction d.

2.2.5. LLE

LLE processes problems from a local aspect. Since the local manifold space can be approximately equivalent to the Euclidean space, LLE recovers the global non-linear geometric structure of the manifold from locally linear fits.

The LLE algorithm includes three steps:

(1) Construct the adjacency graph, and compute the k-nearest neighbors of each point in the graph as well as the first step of ISOMAP;
(2) Utilize these k-nearest neighbor points to compute the matrix of local reconstruction weights, then the matrix and k-nearest neighbor points can be used to compute the output vectors. Reconstruction error is defined as in Eq. (18):

$$\varepsilon(W) = \sum_i |X_i - \sum_j W_{ij} X_j|^2 \tag{18}$$

where $\sum_j w_j = 1$;

(3) Project all the sample features to be low-dimensional, and the condition of the projection is followed as in (19):

$$\min_Y \Phi(Y) = \sum_i |Y_i - \textstyle\sum_j W_{ij} Y_j|^2 \tag{19}$$

which is transformed as in Eq. (20):

$$\Phi(Y)=\sum_{ij} M_{ij}(Y_iY_j) \tag{20}$$

where $M=(1-W)^T(1-W)$, and $\sum_i Y_i=0, \frac{1}{N}\sum_i Y_iY_i^T=I$.

The optimal embedding space can be found by computing the bottom $d+1$ eigenvector of M. As the smallest eigenvalue is close to zero, its minimum eigenvalues from 2 to $d+1$ can be selected to represent the low-dimension.

2.2.6. LPP

LPP is a linear approximation of nonlinear Laplacian Eigenmaps (LE) [26] algorithm. LPP is not only overcome the disadvantages of traditional linear dimensionality reduction methods which are difficult to keep the nonlinear manifold of original data such as PCA, but also to prevent the drawback of non-linear methods which are difficult to get the low-dimensional projected sample points.

LPP has three steps:

(1) Construct the adjacency graph, and compute the k-nearest neighbors of each point in the graph;
(2) Choose the weights, where $W_{ij}=1$, if and only if vertices i and j are connected by an edge;
(3) Compute the eigenvectors and eigenvalues as in (21):

$$XLX^T\boldsymbol{a}=\lambda XD_{dia}X^T\boldsymbol{a} \tag{21}$$

where D_{dia} is a diagonal matrix, $D_{ii}=\sum_j W_{ji} L_p=D_{dia}-W$, L_p is the positive semi-definite Laplacian matrix. According to the constraint condition: $\boldsymbol{a}^TXD_{dia}X^T\boldsymbol{a}=1$, the minimization function can be reduced to find:

$$\arg\min_{\boldsymbol{a}}=\boldsymbol{a}^TXL_pX^T\boldsymbol{a} \tag{22}$$

The eigenvectors which correspond with the top-d smallest non-zero eigenvalues is the projection matrix.

All important notations have been listed in Table 1.

2.3. Construction of image index based on visual vocabulary tree

Vocabulary tree is an efficient data organization structure which can greatly improve the speed of image retrieval. The construction process of a vocabulary tree is shown as Fig. 4.

In practice, we adopt a hierarchical K-means clustering scheme to construct the vocabulary trees of SIFT features and color histogram respectively. Each feature vector is compared with the K clustering center of every layer in a top-down manner to select the closest one until the nearest category is selected. For a vocabulary tree whose height is L, dot product operations in each layer only need K times, so the total number of calculation is KL times. The number of visual words can be calculated by (23):

$$\textstyle\sum_{l=1}^{L} K^l=\frac{K^{L+1}-K}{K-1} \tag{23}$$

In this paper, $K=10$, $L=3$.

2.4. Similarity measurement

The inverted index is used for large-scale image retrieval. We measure the similarity among images using the Term of Frequency–Inverse Document Frequency (TF–IDF).

In the vocabulary tree, each node i is corresponding to a visual word C_i, the term frequency of query image and database images which pass the node i are denoted as q_i and d_i, respectively. The IDF can be computed as in Eq. (24):

$$IDF=\log\frac{N}{N_i}=\omega_i \tag{24}$$

where N is the total number of images, N_i the number of images which pass the node i.

In this paper, $Qi=q_iw_i$ denotes feature vector of query images and $Di=d_iw_i$ denotes feature vector of images from database. The similarity between the query image and those from database can be measured by means of the L_2 norm as in Eq. (25):

$$Similar(D,Q)=\left\|\frac{Q}{||Q||}-\frac{D}{||D||}\right\|_2^2=\sum_N |Q_i-D_i|^2$$

Table 1
List of important notations.

Notations	Meanings	Notations	Meanings
$Sift_{des}[i]x$	horizontal coordinate of i-th SIFT	V^T	liner transformation matrix
$Sift_{des}[i]y$	vertical coordinate of i-th SIFT	S_{inter}	inter-class scatter matrix
T_{opt}	optimization threshold	S_{intra}	intra-class scatter matrix
R_{opt}	optimization range	V_{inter}	local inter-class scatter matrix
N	total number of features	V_{intra}	local intra-class scatter matrix
D	dimension of each eigenvector	$\varepsilon(W)$	reconstruction error
W	projection matrix	W_{ij}	weight
μ	mean of all image features	D_{dia}	diagonal matrix
Cov	covariance matrix	L_p	positive semi-definite Laplacian matrix

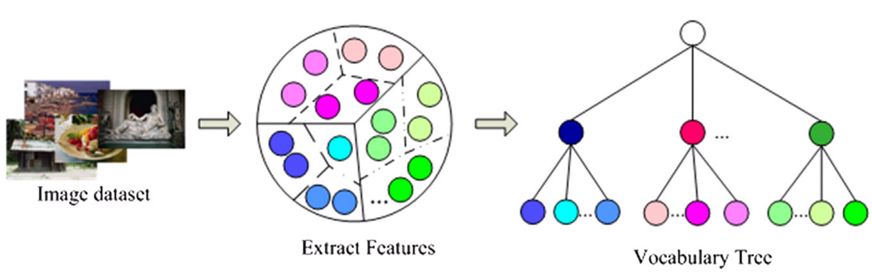

Fig. 4. The process of constructing a vocabulary tree.

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$$
\begin{aligned}
&= \sum_{i|D_i=0} |Q_i|^2 + \sum_{i|Q_i=0} |D_i|^2 + \sum_{i|D_i \neq 0, Q_i \neq 0} |Q_i - D_i|^2 \\
&= 2 - 2 \sum_{i|D_i \neq 0, Q_i \neq 0} Q_i D_i
\end{aligned} \tag{25}
$$

By calculating the sum of products of image elements with corresponding dimension, and selecting specific ones according to similarity results, the proposed algorithm efficiently simplifies the traditional distance measurement method, and thus leads to a significant improvement in retrieval speed.

In the similarity measurement, the dimension of SIFT features and color histogram are reduced and then used to construct two vocabulary trees respectively. According to Eq. (26), the similarity of images is referred and ranked by comparing the index of the query image with those of the images that are stored in the database. The most similar top-k images will be regarded as the research results.

$$Similar = \alpha Sim_{SIFT} + \beta Sim_{HSV} \tag{26}$$

where α, β are the weights of the vocabulary trees of SIFT features and color histogram respectively. By selecting the appropriate weights, it facilitates the image retrieval performance to reach the optimal.

3. Experimental results and analysis

In order to demonstrate the influence of various dimensionality reduction methods mentioned above on the performance of large-scale image retrieval, we performed experiments on an image database containing about 7000 colored images that are chosen from Corel Database [27], image-searching site BaiDu [28], and photo-sharing site Flickr [29]. These images roughly fall into more than 50 categories, such as African people, flowers, airplane, architecture, and so on.

Fig. 5 shows the interface of our proposed large-scale image retrieval system. Initially, 128-dimentional SIFT features and 72-dimentional HSV color histograms are extracted from the images. Then, six different dimensionality reduction methods including PCA, FLDA, LFDA, ISOMAP, LLE, and LPP are used respectively to reduce the dimension of the features. The dimensions of the two kinds of image features are both reduced into 16. And the tuning factor k is set as 7, which will eliminate the irrelevant redundant information of the high dimensional features. Next, two vocabulary trees (branches $K=10$, height level$=3$) are constructed using hierarchical K-means clustering algorithm, which will generate 1110 visual words. The last step is similarity comparison. During this procedure, the weights of α, β are set as $\alpha=1.5$, $\beta=0.3$.

In the experiments, the influence of six dimensionality reduction methods on the retrieval performance is analyzed by comparing the retrieval performance, advantages and disadvantages, computational complexity, and time-consuming as well.

3.1. Retrieval performance

The retrieval performance often adopts precision and recall as the evaluation criteria. In this paper, we define the number of similar images retrieved as *SIR*, the number of non-similar images retrieved as *NSIR*, and the number of similar images not retrieved as *SINR*. The

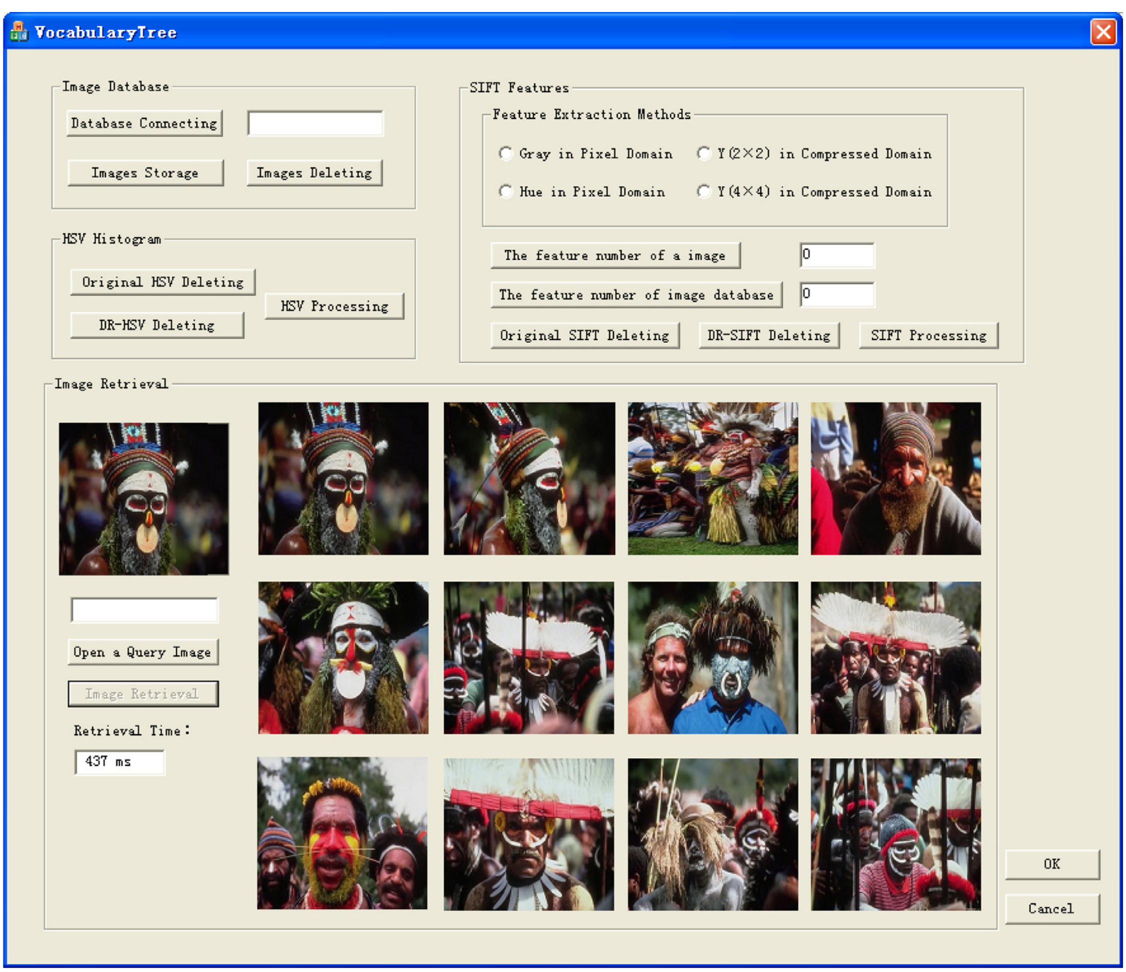

Fig. 5. Large-scale image retrieval interface.

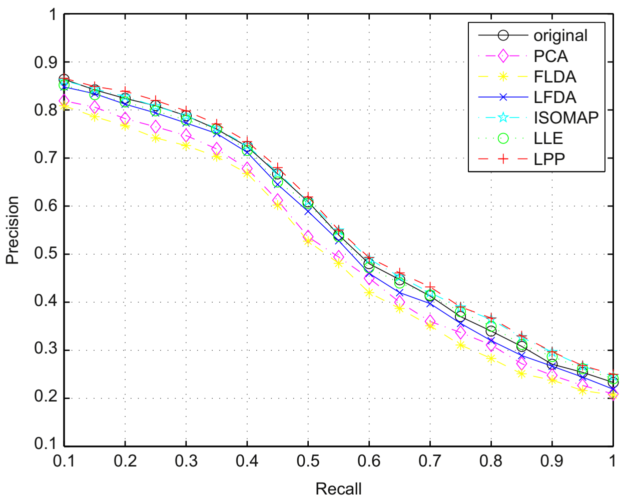

Fig. 6. Recall-precision curves.

definition of precision and recall is shown as in Eqs. (27) and (28):

$$precision = \frac{SIR}{SIR + NSIR} \tag{27}$$

$$recall = \frac{SIR}{SIR + SINR} \tag{28}$$

Precision reflects the accuracy of a retrieval algorithm, while recall reflects the comprehensiveness of the algorithm. The precision-recall curves with six dimension reduction methods are shown in Fig. 6, where the vertical axis is precision ratio and the horizontal axis is recall ratio. As in Fig. 6, seven curves with different colors are shown, representing the recall-precision ratio with original features, and the dimension reduced features using PCA, FLDA, LFDA, ISOMAP, LLE and LPP respectively. And Table 1 shows their precision ratio respectively.

In statistics, the F-measure is another measure of retrieval performance. It considers both the precision and the recall of the test to compute the score. The F_1-measure can be interpreted as a weighted average of the precision and recall, where an F_1-measure reaches its best value at 1 and worst score at 0.

The general F-measure for positive real β is:

$$F_\beta = (1+\beta^2)\frac{precision \cdot recall}{(\beta^2 precision) + recall} \tag{29}$$

The traditional F-measure ($\beta = 1$) is the harmonic mean of precision and recall:

$$F_1 = \frac{2precision \cdot recall}{precision + recall} \tag{30}$$

From Tables 2 and 3, it can be seen that the performance of PCA and LFDA is much worse than that of the other four methods. Since the projection matrix of PCA needs a series of typical images which is only appropriate for the images belonging to this type, the precision ratio of the method using PCA is lower than those using original features. For FLDA, once image features in a particular class have many local means, and FLDA cannot classify the image features correctly. Furthermore, LFDA can keep the retrieval performance close to that of the one using original features.

PCA, LFDA and FLDA are three commonly used linear dimensionality reduction methods which can find a linear transformation matrix, and to reduce the dimension of the features. However, since they are global dimensionality reduction methods, the dimension reduced features cannot reflect the non-linear manifold of the high-dimensional space, and also lose some information of original features, which undermines retrieval performance to a certain degree.

Table 2
The precision ratio of original features and dimensionality reduction methods.

	Original features (%)	PCA (%)	FLDA (%)	LFDA (%)	ISOMAP (%)	LLE (%)	LPP (%)
Precision ratio	86.4	81.9	80.9	84.8	86.1	85.2	86.6

Table 3
The average precision, average recall and F_1-measure using original features and dimensionality reduced features.

	Original features	PCA	FLDA	LFDA	ISOMAP	LLE	LPP
Average precision	0.555	0.514	0.499	0.538	0.563	0.552	0.569
Average recall	0.55	0.55	0.55	0.55	0.55	0.55	0.55
F_1-measure	0.552	0.531	0.523	0.544	0.556	0.551	0.56

The experimental results show that manifold learning dimensionality reduction methods can achieve better retrieval performance than PCA, LFDA and FLDA. However, higher precision ratio achieved by ISOMAP is on the cost of higher computational complexity. And compared with the original features, the dimension reduced features using LPP method can even achieve higher retrieval performance.

3.2. Advantages and disadvantages

Table 4 shows the comparison of the advantages and disadvantages of the six dimensionality reduction methods. It can be seen that PCA has optimal linear reconstruction error with convenient calculation. However, PCA is not always the optimal choice for dimensionality reduction; because the determination of number of principal components has no specific guideline, the dimension of features which has been reduced is not suitable for the classification of inter-class, and will negatively affect the performance of image retrieval to some extent.

FLDA can preserve the discriminative of image features while reducing dimension in the high-dimensional space. Nevertheless, the intra-class scatter is always singular when image features in a particular class have many local means, which prevent it from being corrected.

LFDA could overcome these drawbacks of FLDA, which makes the optimal effect works for both inter-class and intra-class classification. But the size of image vectors used for training cannot much larger than the number of the images. To solve this problem, PCA can be used to reduce the dimension of the input features so that the local intra-class scatter matrix becomes non-singular. LFDA is difficult to keep the nonlinear manifold of original data. Both of FLDA and LFDA are supervised dimensionality reduction methods, they project samples of each class separately in the feature space.

In this case, FLDA and LFDA outperform PCA. Nonetheless, the dimension reduced features by PCA, FLDA and LFDA are difficult to keep the nonlinear manifold of original data.

ISOMAP uses geodesic distance to replace Euclidean distance, and it is better to preserve geometric structure of data. However, ISOMAP is topologically unstable, it requires feature data has the topology of a convex region and cannot solve problems of data having "holes" or "essential loops".

LLE maps its inputs into a single global coordinate system of lower dimensionality. It analyses local symmetries, linear coefficients, and reconstruction errors instead of global constraints, pair-wise distances, and stress functions, which can be invariant to changes of translation, rotation and others. Moreover, optimizations of LLE do not involve local minima.

Table 4
The advantages and disadvantages of dimensionality reduction (DR) methods.

DR methods	Advantages	Disadvantages
PCA	Optimal linear reconstruction error and convenient calculation.	The determination of number of principal components has no specific guideline; Not suitable for the classification of inter-class.
FLDA	FLDA is a supervised linear dimensionality reduction method and preserves the discriminative of image features.	FLDA cannot discriminate images in particular class have many local means. And the intra-class scatter of features is always singular.
LFDA	LFDA makes the optimal effect works for both inter-class and intra-class classification.	Difficult to keep the nonlinear manifold of original data. The size of image vectors used for training cannot be much larger than the number of the images.
ISOMAP	ISOMAP uses geodesic distance to replace Euclidean distance and is better to preserve geometric structure of data.	ISOMAP is topologically unstable and requires feature data having the topology of a convex region.
LLE	LLE can be invariant to changes of translation, rotation and others. Optimizations of LLE do not involve local minima.	LLE requires the manifold of original data that is not closed and is sensitive to noise.
LPP	LPP is easy to get the low-dimensional projected sample points and more focused on the local relation of data.	LPP is difficult to choose the optimal parameter of ε to construct the adjacency graph.

LPP is able to overcome the disadvantages of traditional linear dimensionality reduction methods that are difficult to keep the nonlinear manifold of original data such as PCA. Moreover it can also prevent the drawback of non-linear method which is hard to get the low-dimensional projected sample points. However, LPP is difficult to choose the optimal parameter of ε to construct the adjacency graph.

The existing typical dimensionality reduction methods cannot give comprehensive consideration to any situation. However, by comparing and analyzing these methods, it can be concluded that dimensionality reduction methods of manifold learning (ISOMAP, LLE and LPP) can project high-dimensional of data into a surrogate low-dimensional space more precisely than PCA, FLDA and LFDA. We will further evaluate performance of these dimensionality reduction methods in different aspects as follows.

3.3. Computational complexity

As mentioned above, if the extracted image features are processed directly without dimensionality reduction, it may lead to massive computational complexity. Table 5 shows the computational complexity of six dimensionality reduction methods.

From Table 5, it can be seen that the computational complexity of covariance matrix of PCA and FLDA is $O(DN)$, while the eigenvalue decomposition for the $D\times D$ covariance matrix needs $O(D^3)$. LFDA method needs $O(D^2N)$ to compute the local inter-class scatter matrix V_{inter} and the local intra-class scatter matrix V_{intra}, while the eigenvalue decomposition needs $O(D^3)$. However, in manifold learning, IOSMAP can maintain more information which the computational complexity of computing the $N\times N$ shortest path distance matrix is $O(kN^2\log N)$, and the eigenvalue decomposition needs $O(N^3)$. As a result, ISOMAP require more computations. When the number of data points is increasing, the computational overhead will be unbearable. For LLE, the computational complexity of choosing nearest neighbors is $O(DN^2)$, computing the matrix of local reconstruction weights need $O((D+k)k^2N)$, and solving low-dimensional embedding Y needs $O(dN^2)$. For LPP, the computational complexity of choosing nearest neighbors and computing the matrix of local reconstruction weights are $O(DN^2)$ and $O(dN^2)$ respectively, while solving low-dimensional embedding Y is $O(kDN)$ at most.

Therefore, the computational complexity of PCA, FLDA and LFDA are relatively low, and ISOMAP require more computations. Compared with ISOMAP, the computational complexity of LLE and LPP are obviously lower with less computation requirements. Thus, the execution speed of LLE and LPP are relatively quicker.

3.4. Time-consuming

Table 6 shows the comparison results of time-consuming of query index construction of pre- and post-optimization, and similarity measurement time during the image retrieval process. It can be seen that the method using the pre-optimization features consumes 460 ms, where query index construction and similarity measurement needs 380 ms and 80 ms respectively. The rationale is that using the original high-dimensional features to construct query index based on vocabulary tree spends most of the processing time. Besides, the construction of vocabulary tree is performed offline. The offline training time of features after dimensionality reduction is substantially shortened compared with the time consumption of original high-dimensional features. Moreover, by contrast, as shown in Table 6, after using PCA, FLDA, LFDA, ISOMAP, LEE and LPP dimensionality methods, query index construction time per image has been greatly reduced.

In summary, according to the comparison results of retrieval performance, advantages and disadvantages, computational

Table 5
The computational complexity of dimensionality reduction methods.

Dimensionality reduction (D-R) Methods	Computational complexity
PCA	$O(DN)+O(D^3)$
FLDA	$O(DN)+O(D^3)$
LFDA	$O(D^2N)+O(D^3)$
ISOMAP	$O(kN^2\log N)+O(N^3)$
LLE	$O(DN^2)+O((D+k)k^2N)+O(dN^2)$
LPP	$O(DN^2)+O(kDN)+O(dN^2)$

Table 6
The comparison results of time-consuming of query index construction of pre- and post-optimization, and similarity measurement time during the image retrieval process.

	Comparison of time-consuming		
	D-R methods	***Query index construction time* (ms)**	***Similarity measurement time* (ms)**
pre-optimization	–	380	80
post-optimization	PCA	27	80
	FLDA	27	80
	LFDA	27	80
	ISOMAP	1066	80
	LLE	340	80
	LPP	211	80

Note: The experimental system configuration is Pentium (R) Dual Core CPU Processor at frequency of 3.00 GHz and Memory 1.97 GB, and software including Visual Studio2005, SQL server 2005, and MATLAB R2008a.

complexity, and time-consuming, it indicates that the precision ratio of the methods using LFDA, ISOMAP, LLE and LPP respectively is close to that of the method using the original features. ISOMAP needs the highest computational complexity which will be incapable of affording when the numner of the features increase. Compared with the other methods, LLE and LPP methods need less computational complexity while preserving the higher images retrieval performance, which still enable conducting fast and accurate image retrieval for big database.

4. Conclusions

This paper analyzes and compares the effect of various dimensionality reduction methods (i.e. PCA, FLDA, LFDA, ISOMAP, LLE, and LPP) on the performance of large-scale image retrieval. The experimental results show that the manifold learning methods of LLE and LPP could find the intrinsic geometric constructions or rules and obtain the low-dimensional embedding effectively. The precision ratio and F-measure of LLE and LPP are respectively close and even higher than the method without dimensionality reduction. In addition, LLE and LPP will demonstrably reduce the query index construction time per image. It can be concluded that dimensionality reduction method of LLE and LPP can effectively reduce the dimensions of image features, and maintain the high retrieval performance.

In the future, we will further improve the accuracy of images retrieval by adding the Dominant Color Descriptor (DCD), Color Layout Descriptor (CLD) and Color Quantization Descriptor (CQD) as color information of image.

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